

Enterprise AI Architect Learning Path v3 – Resource Catalog

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How to Use This Catalog

This catalog serves as the master reference for all learning resources used throughout the Enterprise AI Architect Learning Path v3.

Resources are grouped by learning domain rather than provider. Each resource is tagged with:

	☰ Priority	☰ Meaning
1	Core	Must complete
2	Recommended	Strongly encouraged
3	Reference	Use as needed

Domain 1 – Python & AI Engineering Foundations

R0 – 100 Days of Code: The Complete Python Pro Bootcamp

Provider: Udemy

Priority: Core

Why It Matters

Provides the Python foundation needed for AI engineering work.

Focus Areas

- OOP
- Async Programming
- APIs
- File Processing
- Error Handling
- Testing

Primary Phases

- Phase 1

Completion Target

Before Phase 2 begins.

R1 – AI Engineering

Author: Chip Huyen

Provider: O'Reilly

Priority: Core

Why It Matters

Arguably the most important book in the entire learning path.

Provides practical guidance for:

- Building AI applications
- Evaluation
- Prompt engineering
- RAG systems
- Agentic architectures

Primary Phases

- Phase 1
- Phase 2
- Phase 3
- Phase 4
- Phase 6

Suggested Reading Strategy

Read continuously throughout the program rather than finishing it all at once.

R2 – Designing Machine Learning Systems

Author: Chip Huyen

Provider: O'Reilly

Priority: Core

Why It Matters

Introduces production thinking.

Focus Areas

- Monitoring

- MLOps
- Reliability
- Metrics
- Responsible AI

Primary Phases

- Phase 3
 - Phase 7
 - Phase 8
-

Domain 2 – Azure AI Foundations

R3 – Azure AI Foundry Documentation

Provider: Microsoft Learn

Priority: Core

Why It Matters

Primary source of truth for:

- Azure AI Foundry
- Models
- Agents
- Evaluation
- Governance

Primary Phases

- Phase 2
 - Phase 4
 - Phase 7
-

R4 – Azure AI Foundry Training Path

Provider: Microsoft Learn

Priority: Recommended

Why It Matters

Provides guided Foundry learning modules.

Primary Phases

- Phase 2

R19 – AI on Azure: LLMs, Foundational Models & Agents

Provider: Udemy Business

Priority: Recommended

Why It Matters

Provides practical Azure implementation examples.

Primary Phases

- Phase 2
- Phase 7

R20 – Generative AI: Beginner to Pro with OpenAI & Azure OpenAI

Provider: Udemy

Priority: Recommended

Why It Matters

Good implementation-oriented companion course.

Primary Phases

- Phase 2
- Phase 3

Domain 3 – Retrieval-Augmented Generation (RAG)

R5 – Azure AI Search Documentation

Provider: Microsoft Learn

Priority: Core

Why It Matters

Primary reference for:

- Search architecture
- Search indexes
- Query design
- RAG implementations

Primary Phases

- Phase 2
 - Phase 3
-

R6 – Azure AI Search Vector Search Overview

Provider: Microsoft Learn

Priority: Core

Why It Matters

Focuses specifically on:

- Embeddings
- Hybrid Search
- Similarity Search

Primary Phases

- Phase 2
 - Phase 3
-

R7 – Azure AI Search Vector Samples

Provider: GitHub / Microsoft

Priority: Reference

Why It Matters

Provides working code examples.

Primary Phases

- Phase 2
 - Phase 3
-

Domain 4 – Agent Frameworks & Orchestration

R8 – Semantic Kernel Documentation

Provider: Microsoft Learn

Priority: Core

Why It Matters

Microsoft's primary agent framework.

Focus Areas

- Plugins
- Functions
- Filters
- Process Framework

Primary Phases

- Phase 4
 - Phase 5
 - Phase 6
-

R9 – Introduction to Semantic Kernel

Provider: Microsoft Learn

Priority: Recommended

Why It Matters

Provides architectural understanding before diving into implementation.

Primary Phases

- Phase 4
-

R10 – LangGraph Documentation

Provider: LangGraph

Priority: Core

Why It Matters

The most important resource for durable agent workflows.

Focus Areas

- StateGraphs
- Persistence
- Human-in-the-loop workflows

Primary Phases

- Phase 4

- Phase 6
-

R11 – LangGraph with Azure AI Foundry

Provider: Microsoft Learn

Priority: Recommended

Why It Matters

Bridges Azure and LangGraph ecosystems.

Primary Phases

- Phase 4
 - Phase 6
-

R18 – Generative AI with LangChain (2nd Edition)

Provider: O'Reilly

Priority: Core

Why It Matters

Deep treatment of:

- Production agents
- LangGraph
- Advanced RAG
- Workflow patterns

Primary Phases

- Phase 3
 - Phase 4
 - Phase 6
-

R21 – LangChain Academy

Provider: LangChain

Priority: Recommended

Why It Matters

Best hands-on supplement for modern agent development.

Primary Phases

- Phase 4
 - Phase 7
-

Domain 5 – MCP & Tool Ecosystems

R12 – Model Context Protocol Documentation

Provider: MCP

Priority: Core

Why It Matters

Introduction to MCP architecture.

Primary Phases

- Phase 5
-

R13 – MCP Specification

Provider: MCP

Priority: Core

Why It Matters

Defines protocol behavior and tool contracts.

Primary Phases

- Phase 5
-

Domain 6 – LLMOps & Observability

R16 – OpenTelemetry Documentation

Provider: OpenTelemetry

Priority: Core

Why It Matters

Industry standard observability framework.

Primary Phases

- Phase 7
 - Phase 8
-

R17 – .NET Observability with OpenTelemetry

Provider: Microsoft Learn

Priority: Recommended

Why It Matters

Maps AI observability concepts back to familiar .NET patterns.

Primary Phases

- Phase 7
-

Domain 7 – Security & Governance

R14 – OWASP Top 10 for LLM Applications

Provider: OWASP

Priority: Core

Why It Matters

Primary AI security framework for this program.

Primary Phases

- Phase 5
 - Phase 8
-

R15 – NIST AI RMF Generative AI Profile

Provider: NIST

Priority: Core

Why It Matters

Primary governance framework for this program.

Primary Phases

- Phase 8
-

Certification Alignment

AI-900

Status: Completed 

AI-102

Target Window: End of Phase 3 through Phase 7

Resources Most Aligned

- R3
 - R4
 - R5
 - R6
 - R8
 - R19
-

Recommended Study Order

Start Here

1. R0 – Python Bootcamp
2. R1 – AI Engineering

Then

1. R3 – Azure AI Foundry
2. R5 – Azure AI Search
3. R6 – Vector Search

Then

1. R8 – Semantic Kernel
2. R10 – LangGraph
3. R18 – Generative AI with LangChain

Finally

1. R16 – OpenTelemetry

2. R14 – OWASP
3. R15 – NIST AI RMF